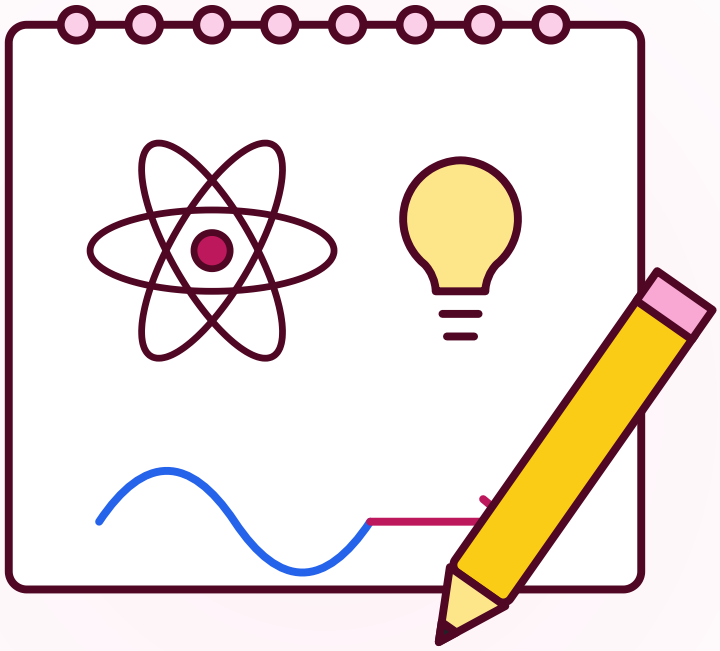
SECRET AGENT

Case File

Draw it fast. Guess it faster. No letters allowed.



AGENT ID

Name: _____

Partner: _____

Event: Science Sketchers

If found, please return to the DDMES Science Olympiad binder shelf.

How a sketch agent works

Science Sketchers is **science Pictionary**. Your team takes turns drawing and guessing science words, and a drawing is a secret message with no letters allowed! Agents who know all 200 words can draw each one in a few seconds. This case file lives in your SciOly binder.

1
READ
Look at the word. What does it really mean?

2
SKETCH
Start with a fast starter picture.

3
POINT
Use an arrow to point to the part that matters.

4
GUESS
Guessers call out lots of related words fast.

5
PASS
Stuck? Pass and move on. No going back.

The Agent's Code

- [ ] No letters, numbers, made-up codes, or hand gestures.
- [ ] The only symbols: arrows, a minus (-) and a plus (+).
- [ ] Nodding yes or no is OK.
- [ ] Everyone draws and everyone guesses.
- [ ] Keep drawings under 10 seconds.
- [ ] Cheer your team, even on a pass!

My agent kit

- [ ] This case file and 2 sharp pencils
- [ ] Scrap paper or a small whiteboard
- [ ] A timer (a phone with an adult)
- [ ] Your flashcards from the word banks
- [ ] Your 5 word banks in your binder
- [ ] Guessers: family and friends

Journal map

Know your event	p. 3-5
Official word list	p. 6-7
Cases #1 to #4	p. 8-11
Sketch pad and log	p. 12-13
Missions and badges	p. 14-15
Answer key (upside down)	p. 16

Where to practice

The dinner table
draw on a napkin while your family guesses

Car rides
a grown-up reads a word, you describe your drawing plan

The weather report
find cold fronts, warm fronts, and hurricanes

Science books
copy the simplest diagram you can find

The backyard
roots, seeds, insects, clouds, and erosion to sketch

The kitchen
melt, freeze, boil, dissolve: before and after!

Did you know? The tournament gives 4 minutes for up to 40 words. To get all 40, your team needs a new word every 6 seconds!

Mission briefing: the event

Teams of 2 or 3 take turns drawing and guessing up to 40 science words from the official list in 4 minutes, with no letters, numbers, or gestures; they bring only a pencil and may look at the word list just before entering the room.

The rules in one box

Time	4 minutes to guess up to 40 words. Every team gets the same words in the same order.
Take turns	After each word, the next teammate draws. Everyone draws and guesses.
Drawer rules	No letters, numbers, codes, or hand gestures (a rule break makes the word a pass). Nodding yes or no is OK.
Allowed symbols	Arrows. A - to shorten a word, a + to lengthen it (refract + = refraction).
Pass	Anyone can pass. You cannot go back to a passed word.
Scoring	1 point per word. Plurals count; other word forms do not.

Scoring

- ☐ 1 point for each word your team guesses.
- ☐ Plurals count (cells for Cell).
- ☐ Other forms do not count (evaporate is not Evaporation). Use + or -!
- ☐ **Tie?** The team with the longest streak of correct words in a row wins.

Before you walk in

- ☐ Bring only a pencil.
- ☐ You may look at the word list just before you enter the room.
- ☐ Teams have 2 or 3 agents.
- ☐ Every team gets the same words in the same order.

How a turn works

1 The drawer looks at the word. Only the drawer!

2 Draw fast. Guessers shout words.

3 Nod when they are close. Use + or - for almost-right words.

4 Got it? The next teammate draws the next word.

5 Stuck? Anyone can say pass.

Our drawing order

1st _____ 2nd _____ 3rd _____

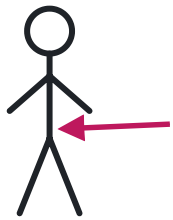
Pass fast. If nobody is close after about 15 seconds, pass. Four minutes goes by quickly, and there are 40 words to try.

Drawing rules and secret symbols

A drawer may use only 3 symbols: **arrows**, a **minus (-)**, and a **plus (+)**. Anything else (a letter, a number, a code, or a hand signal) turns the word into a pass.

Allowed

✓ Arrow: point to the part



✓ Arrow: show a change



✓ + makes the word longer



Using + and -

Team says "evaporate" but the card says **Evaporation**? Draw a **+**: make it longer.

Team says "erosion" but the card says **Erode**? Draw a **-**: make it shorter.

Two-word terms

Draw both parts and join them. **Sound wave:** an ear plus a wavy line. **Food web:** animals with arrows from each food to its eater.

Not allowed

✗ Letters



✗ Numbers



✗ Hand gestures



Thermometers, scales, rulers, clocks, and graphs get **no numbers**. Weather maps get no H or L: draw a front with triangles or half circles instead.

Try it: arrows only

Revolve

Rotation

Repel

Starter pictures

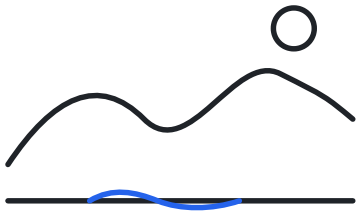
Most words start from one of 6 fast pictures. Draw the starter picture in 2 seconds, then add an **arrow** to the part you mean.

A person



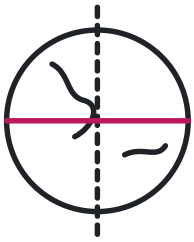
Use it for: Joint, Muscle, Stomach, Spinal cord, Digestive system, and every scientist (person + tool)

A landscape



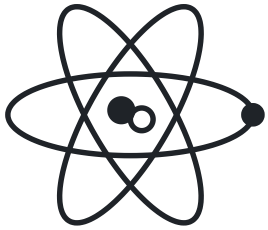
Use it for: Erode, River, Lake, Elevation, Estuary, Soil, Earthquake, Water table

The Earth



Use it for: Equator, Arctic, Pole, Season, Rotation, Orbit, Revolve, Atmosphere

An atom



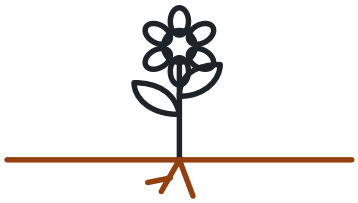
Use it for: Proton, Neutron, Electron, Molecule, Compound, Charge

The sky



Use it for: Cloud, Precipitation, Condensation, Humidity, Air pressure, Jet stream

A plant



Use it for: Root, Stem, Leaf, Flower, Pollen, Seed, Germinate, Photosynthesis

More agent tricks

Scientists	Person + tool. Astronomer: telescope. Geologist: rock hammer. Entomologist: net and bug. Botanist: plant and hand lens. Chemist: bubbling flask.
Joint	A bent arm with a circle at the elbow.
Digestive system	Person outline with a tube from mouth to stomach to a curly intestine.
Atom	Dots in the middle with rings around them.
Compound vs mixture	Compound: two different balls stuck together. Mixture: a bowl of trail mix.
Before and after	Melt: ice cube, arrow, puddle. Expand: small balloon, arrow, big balloon.

The official word list

Any of these 200 words can come up at the tournament. Check a word when you can draw it in **10 seconds** and a teammate guesses it. Check all 200 for the **Word List Master** badge.

Bank 1: Living things and ecosystems (40 words)

- ☐ Adapt
- ☐ Amphibian
- ☐ Antenna
- ☐ Bird
- ☐ Camouflage
- ☐ Carnivore
- ☐ Consumer
- ☐ Decay
- ☐ Decomposer
- ☐ Desert
- ☐ Egg
- ☐ Environment
- ☐ Exoskeleton
- ☐ Extinct
- ☐ Flower
- ☐ Food web
- ☐ Forest
- ☐ Fungus
- ☐ Germinate
- ☐ Habitat
- ☐ Herbivore
- ☐ Insect
- ☐ Leaf
- ☐ Mammal
- ☐ Metamorphosis
- ☐ Mutualism
- ☐ Omnivore
- ☐ Organism
- ☐ Parasite
- ☐ Photosynthesis
- ☐ Poison
- ☐ Pollen
- ☐ Population
- ☐ Predator
- ☐ Prey
- ☐ Producer
- ☐ Reptile
- ☐ Root
- ☐ Seed
- ☐ Stem

Bank 2: Body, cells, atoms, and scientists (40 words)

- ☐ Abdomen
- ☐ Astronomer
- ☐ Atom
- ☐ Biologist
- ☐ Blood
- ☐ Bone
- ☐ Botanist
- ☐ Calorie
- ☐ Cell
- ☐ Chemist
- ☐ Chromosome
- ☐ Compound
- ☐ Data Scientist
- ☐ Digestive system
- ☐ Eardrum
- ☐ Electron
- ☐ Endoskeleton
- ☐ Engineer
- ☐ Entomologist
- ☐ Exercise
- ☐ Geologist
- ☐ Joint
- ☐ Mixture
- ☐ Molecule
- ☐ Muscle
- ☐ Neutron
- ☐ Organ
- ☐ Oxygen
- ☐ Physicist
- ☐ Protein
- ☐ Proton
- ☐ Saliva
- ☐ Skin
- ☐ Spinal cord
- ☐ Spine
- ☐ Stomach
- ☐ Sugar
- ☐ Teeth
- ☐ Vitamin
- ☐ Waste

Tip: quiz yourself in the order your family reads them, not the order on this page. The tournament shuffles the words.

The official word list, continued

Bank 3: Earth, water, and weather (40 words)

- ☐ Acid rain
- ☐ Air pressure
- ☐ Arctic
- ☐ Atmosphere
- ☐ Barometer
- ☐ Climate
- ☐ Cloud
- ☐ Cold front
- ☐ Condensation
- ☐ Cumulus Cloud
- ☐ Earthquake
- ☐ Elevation
- ☐ Equator
- ☐ Erode
- ☐ Estuary
- ☐ Evaporation
- ☐ Fossil
- ☐ Global warming
- ☐ Humidity
- ☐ Humus
- ☐ Hurricane
- ☐ Igneous rock
- ☐ Jet stream
- ☐ Lake
- ☐ Map
- ☐ Metamorphic rock
- ☐ Mineral
- ☐ Ocean
- ☐ Pollution
- ☐ Porous
- ☐ Precipitation
- ☐ Quartz
- ☐ Recycle
- ☐ River
- ☐ Sedimentary rock
- ☐ Soil
- ☐ Tornado
- ☐ Warm front
- ☐ Water table
- ☐ Weather

Bank 4: Space and energy (40 words)

- ☐ Charge
- ☐ Circuit
- ☐ Comet
- ☐ Conduction
- ☐ Constellation
- ☐ Crater
- ☐ Current
- ☐ Eclipse
- ☐ Electricity
- ☐ Galaxy
- ☐ Gravity
- ☐ Heat
- ☐ Insulate
- ☐ Light
- ☐ Lightning
- ☐ Magnet
- ☐ Meteor
- ☐ Moon
- ☐ Orbit
- ☐ Planet
- ☐ Pole
- ☐ Radiation
- ☐ Repel
- ☐ Revolve
- ☐ Rocket
- ☐ Rotation
- ☐ Season
- ☐ Shadow
- ☐ Sound wave
- ☐ Space
- ☐ Star
- ☐ Static Electricity
- ☐ Sun
- ☐ Sunrise
- ☐ Sunset
- ☐ Telescope
- ☐ Thunder
- ☐ Vibrate
- ☐ Waning Moon
- ☐ Waxing Moon

Bank 5: Matter, force, motion, and measuring (40 words)

- ☐ Absorb
- ☐ Angle
- ☐ Balance
- ☐ Boil
- ☐ Clockwise
- ☐ Compass
- ☐ Compression
- ☐ Contract
- ☐ Degree
- ☐ Dissolve
- ☐ Energy
- ☐ Estimate
- ☐ Expand
- ☐ Flight
- ☐ Float
- ☐ Force
- ☐ Freeze
- ☐ Friction
- ☐ Gas
- ☐ Gram
- ☐ Graph
- ☐ Ice
- ☐ Liquid
- ☐ Mass
- ☐ Matter
- ☐ Measure
- ☐ Melt
- ☐ Meter
- ☐ Motion
- ☐ Pendulum
- ☐ Plastic
- ☐ Pull
- ☐ Quantity
- ☐ Ramp
- ☐ Scale
- ☐ Solid
- ☐ Temperature
- ☐ Thermometer
- ☐ Velocity
- ☐ Volume

Words checked: _____ / 200 **Date I finished:** _____

My slowest words (practice these on the Sketch pad, p. 12)

Word	My fast drawing idea	Word	My fast drawing idea

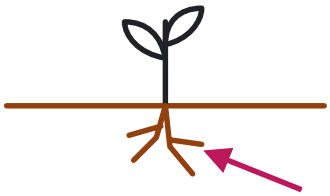
Case #1: Mystery Sketches

TOP SECRET

ASSIGNED after Lesson 1 (Fri Oct 16) · **BADGE** **Life Sketcher**

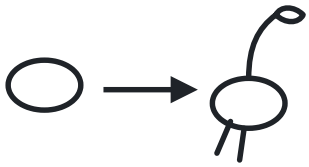
YOUR MISSION Another agent left these 6 sketches. Every one is a word from Bank 1 (living things). Crack them all, then draw 5 Bank 1 words in 10 seconds each while a family member guesses.

Sketch A



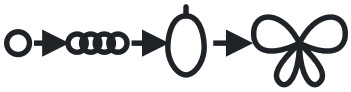
Word: _____

Sketch B



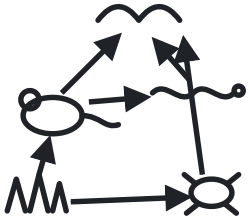
Word: _____

Sketch C



Word: _____

Sketch D



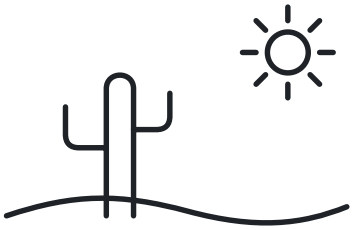
Word: _____

Sketch E



Word: _____

Sketch F



Word: _____

Check your answers on p. 16.

Family guessing game

#	Word I drew	Guessed in 10 sec?	What they guessed first
1			
2			
3			
4			
5			

My hardest word, and a better drawing idea:

Case #2: The Scientist Line-up

TOP SECRET

ASSIGNED after Lesson 2 (Fri Nov 20) · **BADGE** **Body Sketcher**

YOUR MISSION Draw a scientist as a **person + their tool**. Match each scientist to the tool you would draw. Then sketch 3 body words with a starter picture and an arrow. Make 10 flashcards from Bank 2 too.

Astronomer	●	● rock hammer
Geologist	●	● net and a bug
Entomologist	●	● bubbling flask
Botanist	●	● telescope
Chemist	●	● hard hat and a bridge
Engineer	●	● a graph with dots (no numbers!)
Data Scientist	●	● plant and a hand lens

Starter picture + arrow

Joint

Eardrum

Spinal cord

Flashcard tracker

Make 10 flashcards from Bank 2: word on the front, your 10-second drawing on the back. Check a box each time a family member guesses it.

Word	Try 1	Try 2	Word	Try 1	Try 2

All 10 guessed twice? Show a coach your flashcards for the **Body Sketcher** badge.

Case #3: Plus or Minus?

TOP SECRET

ASSIGNED after Lesson 3 (Fri Jan 15) · **BADGE** Earth Sketcher

YOUR MISSION With a grown-up, look at a weather map on TV or online and find a cold front or a warm front. Sketch it without letters. Then crack the Plus or minus puzzle (get all 8 for the **Plus-Minus Pro** badge).

Weather map hunt

Sketch the front I found (no H, no L, no numbers!)

Date: _____ Where I saw it: _____

Cold front: a line with triangles. Warm front: a line with half circles.

☐ cold front ☐ warm front ☐ both

Plus or minus?

Your team shouts the word in quotes, but the card says the bold word. Do you draw + (longer) or - (shorter)?

#	Card	Team says	Draw
1	**Evaporation**	“evaporate”	◯ + ◯ -
2	**Erode**	“erosion”	◯ + ◯ -
3	**Condensation**	“condense”	◯ + ◯ -
4	**Pollution**	“pollute”	◯ + ◯ -
5	**Recycle**	“recycling”	◯ + ◯ -
6	**Vibrate**	“vibration”	◯ + ◯ -
7	**Rotation**	“rotate”	◯ + ◯ -
8	**Germinate**	“germination”	◯ + ◯ -

Plan it: fast ideas for tricky Earth words

Water table

Estuary

Porous

Jet stream

Case #4: The Home Tournament

TOP SECRET

ASSIGNED after Lesson 4 (Fri Feb 12) · **BADGE** **Speed Sketcher**

YOUR MISSION Run a tournament at home with your family. A grown-up holds this page and shows the drawer one word at a time. Play 2 rounds of 2 minutes each. Mark C (correct) or P (passed). Then review the full word list the morning of the tournament.

#	Word	Rd 1	Rd 2	#	Word	Rd 1	Rd 2
1	**Mammal**			11	**Insect**		
2	**Eclipse**			12	**Planet**		
3	**Thermometer**			13	**Teeth**		
4	**Spine**			14	**Hurricane**		
5	**River**			15	**Balance**		
6	**Seed**			16	**Parasite**		
7	**Circuit**			17	**Rocket**		
8	**Astronomer**			18	**Atom**		
9	**Cold front**			19	**Soil**		
10	**Melt**			20	**Float**		

Our results

Round	Correct	Passed	Longest streak	Words to practice
1				
2				

Goal for the **Speed Sketcher** badge: 10 or more correct in one 2-minute round.

Our drawing order for the tournament, and why:

Tournament morning (Sat Mar 13)

- ☐ Read the whole word list once (p. 6-7).
- ☐ Practice my 5 slowest words.
- ☐ 1 sharp pencil. Nothing else!
- ☐ Look at the word list right before we go in.

Sketch pad

Practice your weak words here. Write the word under each box, set a 10-second timer, and draw. Draw 10 Bank 4 words (space and energy) for the **Space Sketcher** badge. Draw 10 words with no letters or numbers, checked by a grown-up, for **Rule Keeper**.

Word:	Word:	Word:	Word:
Word:	Word:	Word:	Word:
Word:	Word:	Word:	Word:
Word:	Word:	Word:	Word:
Word:	Word:	Word:	Word:

Grown-up check: no letters, numbers, or codes on this page. Initials: _____

Evidence: Speed Round Log

Log every practice round: at meetings, in labs, and at home. A **streak** is how many words in a row your team got right with no pass. Get a streak of 5 for the **Streak Star** badge.

Date	Time (min)	Words tried	Correct	Passed	Longest streak	Words to practice

My best round: _____ correct

My longest streak: _____

Case Missions

One case for each Science Sketchers meeting. When you finish, show a coach or your Event Captain for initials, then color in the badge.

When	Mission	Badge	Done	Initials
After Fri Oct 16 Lesson 1	Case #1: crack the 6 Mystery Sketches and play the family game (p. 8).	Life Sketcher	☐	
After Fri Nov 20 Lesson 2	Case #2: The Scientist Line-up and 10 flashcards (p. 9).	Body Sketcher	☐	
After Fri Jan 15 Lesson 3	Case #3: find a real front (p. 10). Get all 8 Plus or minus right.	Earth Sketcher / Plus-Minus Pro	☐	
After Fri Feb 12 Lesson 4	Case #4: The Home Tournament, 10+ correct in one round (p. 11).	Speed Sketcher	☐	
Anytime Bonus	Sketch pad: 10 space and energy words, and 10 rule-perfect drawings (p. 12).	Space Sketcher / Rule Keeper	☐	
Anytime Bonus	A streak of 5 in the Speed Round Log (p. 13).	Streak Star	☐	
Anytime Bonus	Check all 200 words on the official list (p. 6-7).	Word List Master	☐	

Tournament day: Sat Mar 13

- ☐ 1 sharp pencil (that is all you may bring)
- ☐ Look at the word list just before you go in
- ☐ A good night's sleep

- **Draw fast:** starter picture, then an arrow.
- Use + and - when they are almost right.
- Pass after about 15 seconds. No going back.

CASE CLOSED

This agent earned all 9 badges and is now a **Master Sketcher**.

Agent _____ Coach or Event Captain _____ Date _____

My Badges

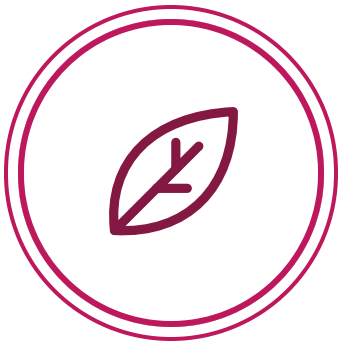
Color in each badge when you earn it. Earn all 9 to become a **Master Sketcher**! Your Team Badge Passport has the same badges.



Rule Keeper

10 drawings, no letters or numbers

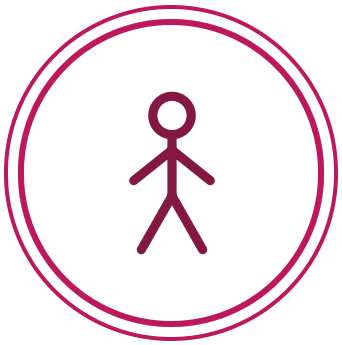
Initials: _____



Life Sketcher

Case #1: Mystery Sketches

Initials: _____



Body Sketcher

Case #2: Scientist Line-up

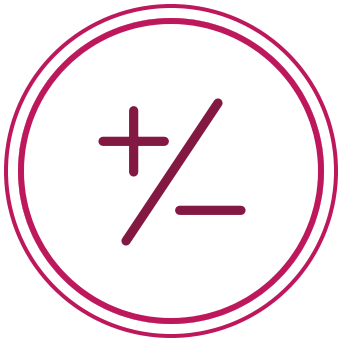
Initials: _____



Earth Sketcher

Case #3: find a real front

Initials: _____



Plus-Minus Pro

All 8 Plus or minus right

Initials: _____



Space Sketcher

10 space and energy sketches

Initials: _____



Speed Sketcher

Case #4: Home Tournament

Initials: _____



Streak Star

A streak of 5 in a row

Initials: _____



Word List Master

Check all 200 words

Initials: _____

Answer key

No peeking until you have tried the page!

Case #1: Mystery Sketches (p. 8)

A: Root. B: Germinate. C: Metamorphosis. D: Food web. E: Decomposer. F: Desert.

Case #2: Scientist Line-up (p. 9)

Astronomer: telescope. Geologist: rock hammer. Entomologist: net and a bug. Botanist: plant and a hand lens. Chemist: bubbling flask. Engineer: hard hat and a bridge. Data Scientist: a graph with dots.

Joint: a bent arm with a circle at the elbow. Eardrum: a head with an arrow into the ear. Spinal cord: a person from the side with an arrow to the back.

Case #3: Plus or minus? (p. 10)

1. Evaporation: +. 2. Erode: -. 3. Condensation: +. 4. Pollution: +. 5. Recycle: -. 6. Vibrate: -. 7. Rotation: +. 8. Germinate: -.

Water table: ground cut open with a line of water underground and a well going down to it. Estuary: a river flowing into the ocean, with marsh grass. Porous: a rock or sponge full of holes, with drops going in. Jet stream: the Earth with a fast wavy arrow high above it.

Scoring reminders

1 point per word. Plurals count; other word forms do not. Ties go to the longest streak of correct words in a row.